QSAR-X

Target-specific QSAR workflow, training, validation, and deployment report

Scope

This report documents the current local QSAR-X project from raw assay retrieval through the web application. It covers the 20-target kinase panel, the EGFR model in detail, model artifacts, validation evidence, prediction output, and the present release status.

Current release state: 20 individual target-specific models are registered and prediction-enabled for screening. No model is labelled externally validated because the strict evidence gates have not yet all passed. This distinction is deliberate and prevents overstatement of model performance.

Executive inventory

Item	Current value	Evidence / location
Target models	20	models/target_registry.json
Curated panel rows	87,095	data/common/kinase_panel_ic50.csv
Endpoint	IC50 normalized to pIC50	pIC50 = 9 - log10(IC50 nM)
Chemical representation	Morgan fingerprint, radius 2, 2048 bits	RDKit feature generation
Selected model family	Extra Trees candidate, target-specific	300 estimators; per-target training report
Prediction interfaces	Single SMILES and batch CSV	FastAPI backend and Vite frontend

1. End-to-end workflow

1. Target panel definition

A 20-member human kinase panel was defined with ChEMBL target identifiers, target metadata, UniProt accession, and local FASTA sequence files.

2. Assay retrieval

The ChEMBL activity service was queried per target. IC50 records were retained only when the target was a direct single protein, the activity was numeric and positive, units were nM, and the relation was exact (=). Raw and cleaned target datasets were stored locally.

3. Curation and standardization

SMILES were canonicalized with RDKit. Measurements were converted to pIC50. Duplicate structures were aggregated using the median activity. Target-specific datasets are retained in data/targets/CHEMBL*/IC50_clean.csv.

4. Feature engineering

Each compound is represented by an RDKit Morgan circular fingerprint (radius 2; 2,048 bits). The individual target models use this chemical representation. Protein sequences and broader multitarget experimentation are retained separately but do not replace target-specific models.

5. Training and selection

For every target, chirality-aware Bemis-Murcko scaffolds were used to separate training, validation, and scaffold-held-out test compounds. Median baseline, Extra Trees, Random Forest, and SVR were compared by training CV RMSE. The selected candidate is stored with its version, split assignments, evaluation report, and serialized model.

6. External checks

BindingDB REST records were pulled using the target UniProt accession. Exact chemical overlap with the ChEMBL training set was removed. The reports are kept as external-source evidence; they do not automatically promote any model because BindingDB REST omits relation operators and complete assay provenance.

7. Independent PubChem search

PubChem BioAssay PUG-REST was used to discover non-ChEMBL IC50 assays. Candidate records must provide explicit IC50, unit, exact relation, SMILES, assay context and PubMed provenance before ingestion. The JAK2 AID 1699 extraction produced 105 rows; 95 overlapped training structures, leaving 10 independent structures, below the 30-compound minimum gate.

8. Deployment

The FastAPI service loads the versioned registry; the Vite frontend provides target selection, SMILES prediction, batch CSV input, CSV exports, and browser-local history export.

2. Data sources, downloaded artifacts, and traceability

The system uses public scientific sources. Data were saved as local immutable-ish project artifacts with target metadata, raw activities, curated activities, protein sequence, model evaluation, and external evidence files. The workflow never creates IC50 values synthetically.

Source	Use in QSAR-X	Stored evidence
ChEMBL	Training IC50 assay data and target metadata for each target.	data/targets/CHEMBL*/target_metadata.json; IC50_raw.csv; IC50_clean.csv
UniProt via target metadata	Protein accession and FASTA identity for target resolution.	data/targets/CHEMBL*/protein.fasta
BindingDB REST	External-source non-overlap check by UniProt accession.	models/CHEMBL*/baseline_v*/external_validation_report.json
PubChem BioAssay PUG-REST	Discovery and curation of independent IC50 assay candidates.	data/external_validation/pubchem_assay_inventory.json; pubchem_independent/

Curation rules

The training path accepts IC50 endpoint values in nM with an exact relation and a positive numeric value. pIC50 is calculated as 9 - log10(IC50 nM). Canonical SMILES make duplicate detection and overlap removal structure-aware. The report records source hashes in individual evaluation reports so an artifact can be linked to its training dataset and model file.

Why external data cannot be reused blindly

ChEMBL values are the training data, so they cannot serve as an independent test. BindingDB and PubChem are evaluated separately. Exact structures present in a training dataset are removed before external metrics are calculated. Assay endpoint, unit, relation, target accession, assay context, and publication provenance are checked before a validation result can be promoted.

3. EGFR model dossier

EGFR target: CHEMBL203 | UniProt: P00533 | Endpoint: IC50

Training data and split

Measure	EGFR value
Curated compounds	10834
Unique scaffolds	3977
Split counts	Train 7023; validation 1443; scaffold test 2368
Fingerprint	Morgan radius 2, 2,048 bits
Selected candidate	ExtraTrees with 300 estimators

EGFR internal model comparison

Model	CV R2	CV RMSE	CV MAE
MedianBaseline	-0.008	1.380	1.143
ExtraTrees	0.569	0.902	0.681
RandomForest	0.561	0.911	0.693
SVR	0.563	0.907	0.689

EGFR selected candidate performance

Evaluation	R2	RMSE pIC50	MAE pIC50
Validation split	0.599	0.851	0.613
Scaffold-held-out test	0.552	0.850	0.635

EGFR external-source evidence

Measure	Value
BindingDB exact-structure overlaps removed	1414
Remaining external non-overlap compounds	770
External RMSE / MAE / Spearman	1.665 / 1.390 / 0.366
Quality-gate result	Not passed - relation and assay context are absent in the BindingDB REST response; external metrics are also weak.

Interpretation

EGFR has an internally scaffold-held-out candidate and a documented BindingDB external-source check. The external check is intentionally not labelled a validated model: its RMSE is 1.665 pIC50 and the source does not provide the complete relation and assay-context fields required by the local gate. The UI therefore exposes this model for screening with an explicit model status, not as commercial validation.

4. 20-target model inventory

Each listed target has an individual baseline_v1 or later artifact. “External record” means an external-source report exists. It does not mean that the strict validation gates were passed.

Gene	ChEMBL ID	n	Scaffold RMSE	External n	External RMSE	Gate
ABL1	CHEMBL1862	2151	0.683	139	1.146	Not passed
AKT1	CHEMBL4282	3455	0.731	91	0.792	Not passed
BRAF	CHEMBL5145	5384	0.656	203	1.228	Not passed
CDK2	CHEMBL301	2115	0.843	741	1.040	Not passed
EGFR	CHEMBL203	10834	0.850	770	1.665	Not passed
ERBB2	CHEMBL1824	2522	0.741	1497	1.404	Not passed
FGFR1	CHEMBL3650	3596	0.657	103	0.749	Not passed
FLT3	CHEMBL1974	3766	0.782	52	0.861	Not passed
IGF1R	CHEMBL1957	2324	0.593	33	0.727	Not passed
JAK2	CHEMBL2971	10405	0.696	10	0.333	Not passed
KDR	CHEMBL279	9762	0.759	160	0.829	Not passed
KIT	CHEMBL1936	1946	0.750	128	1.010	Not passed
MAP2K1	CHEMBL3587	866	0.767	63	0.751	Not passed
MAPK14	CHEMBL260	4744	0.720	245	1.148	Not passed
MET	CHEMBL3717	4193	0.647	83	0.813	Not passed
MTOR	CHEMBL2842	4354	0.603	28	0.707	Not passed
PDGFRA	CHEMBL2007	632	0.798	71	1.073	Not passed
PIK3CA	CHEMBL4005	6465	0.616	566	1.202	Not passed
RET	CHEMBL2041	3980	0.612	95	1.413	Not passed
SRC	CHEMBL267	3601	0.841	62	0.638	Not passed

Model availability in the application

All 20 models are registered in models/target_registry.json and are selectable in the frontend. A prediction response contains canonical SMILES, pIC50, IC50 nM, a dataset-level error estimate, nearest training similarity, model version, and a screening confidence label. Batch predictions accept a CSV with a smiles column and can be exported as a CSV. Single predictions and browser history can also be exported as CSV.

5. Validation protocol and present status

Automated quality gates

Gate	Requirement
Target identity	External target accession must match the expected UniProt accession.
Endpoint and units	The dataset must be IC50 and normalized to nM.
Relation	Only exact = values can enter the strict external validator.
Assay context and publication	Assay context and a publication identifier must be documented.
Independence	Exact training-structure overlaps are removed; publication overlap is also checked when provenance is present.
Sample size	At least 30 independent external compounds are required.
Performance floor	RMSE <= 1.0 pIC50, MAE <= 0.75 pIC50, Spearman >= 0.50.
Reproducibility	Model hash must match the candidate hash recorded at training time.

Current status

No model currently passes every strict external validation gate. This is not a training failure: all 20 target-specific artifacts were trained and registered. It means the current independent evidence is incomplete, has missing source fields, insufficient remaining non-overlap compounds, or does not meet the predefined metric thresholds. The release must be described as screening models with recorded external evidence, rather than as commercially validated predictors.

Recommended route to validated status

For each target: collect an assay-level independent dataset with IC50, exact relation, nM units, target accession, documented protocol/context and publication ID; exclude all training structures and publication overlaps; retain at least 30 compounds; run the versioned external validator; inspect the prediction residuals and source provenance; then conduct an expert review before changing the registry status. A prospective wet-lab test remains the strongest production claim.

6. Software and reproducibility

Backend: FastAPI. Frontend: React/Vite. Chemical processing: RDKit. Model training: scikit-learn Extra Trees, Random Forest, and SVR comparison. Python runtime recorded in training reports: 3.12.14; RDKit 2026.09.1; scikit-learn 1.9.1. Each candidate folder stores candidate_model.joblib, evaluation_report.json, model comparison CSV, and split assignments.

Project entry points

scripts/build_kinase_panel.py builds the panel. scripts/train_panel_targets.py trains the individual targets. scripts/validate_bindingdb_target.py creates non-promotional BindingDB evidence reports. scripts/discover_pubchem_ic50_assays.py discovers PubChem candidate assays. scripts/extract_pubchem_independent_ic50.py filters eligible independent PubChem records. scripts/validate_external_target.py applies strict external quality gates. scripts/sync_target_registry.py registers the local target candidates for the API.

End of report